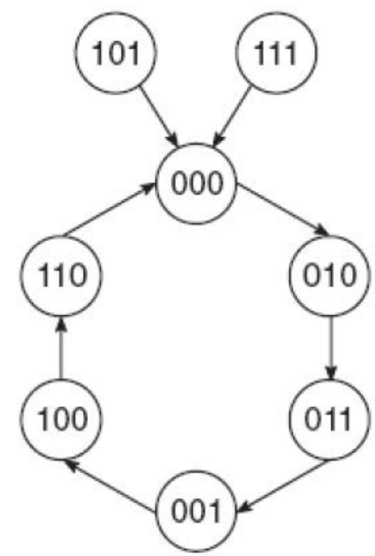
United International University
QUEST FOR EXCELLENCE

CSE 1326: Digital Logic Design Lab
Counters with Arbitrary Sequences

United International University

Objective

- Implement an arbitrary sequence Counter using D and J-K flip-flop.
  - Count the following sequence repeatedly
    - 2->3->1->4->6->0->2->so on, Sec D, E, no class after in class assignment.
  - If accidentally the counter starts with ‘5’ or ‘7’, it should move to ‘0’ in the next step



State Diagram

Flip-flop input table: using J-K

Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

J-K Excitation Table

2->3->1->4->6->0->2->so on

Present state			Next state			Inputs					
Q_2	Q_1	Q_0	D_2 Q_2	D_1 Q_1	D_0 Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	1	0	0	X	1	X	0	X
0	0	1	1	0	0	1	X	0	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	0	0	1	0	X	X	1	X	0
1	0	0	1	1	0	X	0	1	X	0	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	0	0	0	X	1	X	1	0	X
1	1	1	0	0	0	X	1	X	1	X	1

Input Equations

$J_2 = Q_0 \,.\, \overline{Q_1}$

$K_2 = Q_1 + Q_0$

$J_1 = \overline{Q_0}$

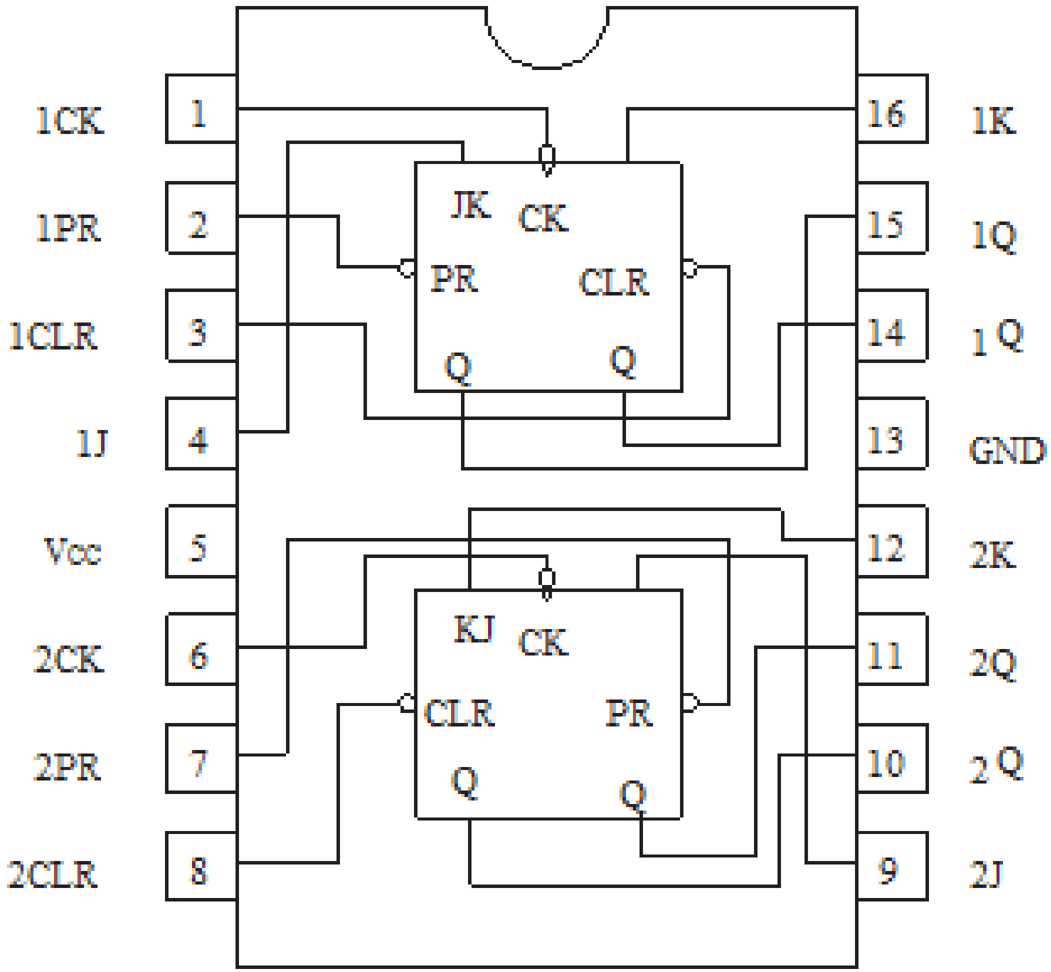
$K_1 = Q_2 + Q_0$

$J_0 = Q_1 \,.\, \overline{Q_2}$

$K_0 = \overline{Q_1} + Q_2$

Verify that you also have the same equations.

74LS76AP: Dual J-K Flip-Flops (with Preset and Clear)



Writing Report

- ICs being used
- For the arbitrary sequence counter, provide for both J-K and D flip-flop
  - Flip-flop Input/Output (state) tables
  - K-maps
  - Equations
  - Circuit diagrams
- Answer the following questions:
  - How can you test the statement, “If the counter starts with ‘5’ or ‘7’, it should move to ‘0’ ?”